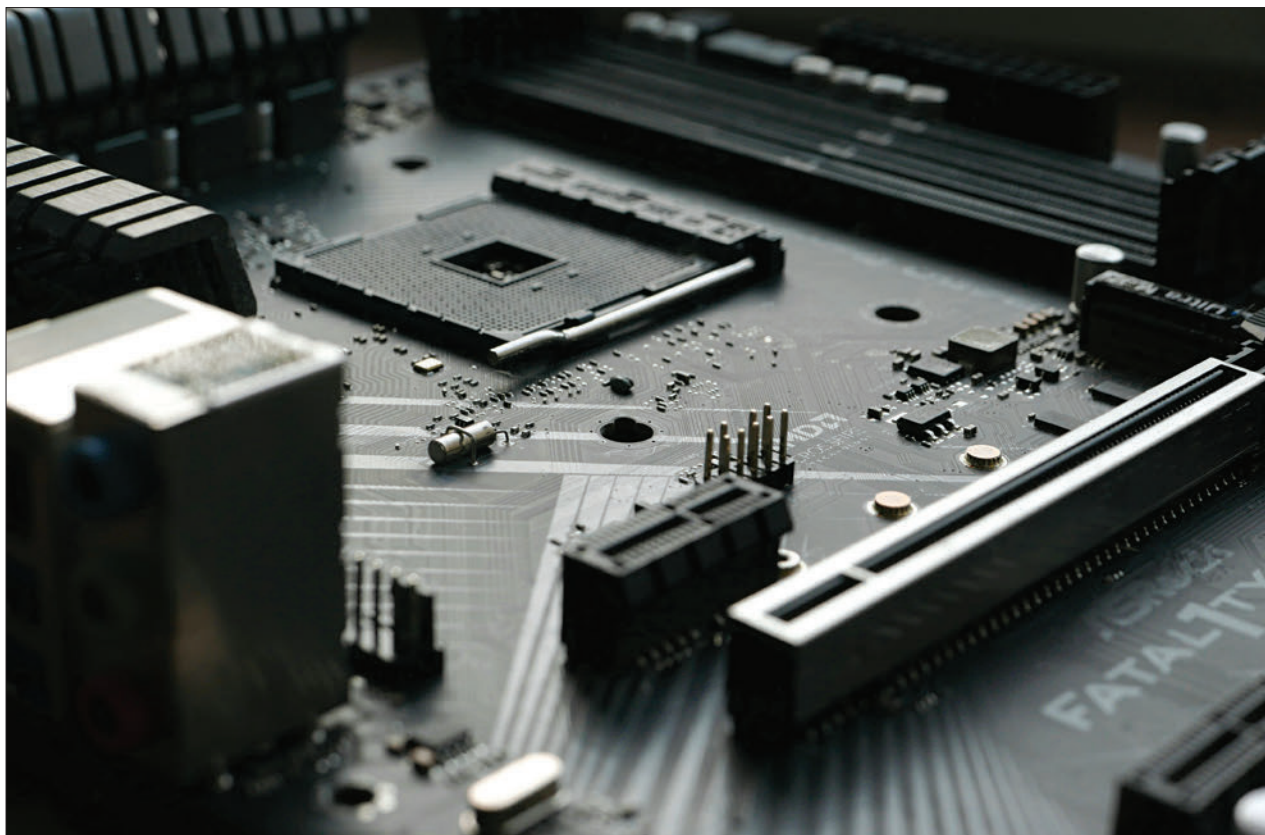




The Nerve Centre – Motherboards

Balancing chipsets, form factors and power delivery effectively

The motherboard is the backbone of your PC: it supplies power, interconnects the CPU, graphics card and memory, and exposes headers for storage and front-panel ports. Picking the right board influences everything from the processor you can use to how many NVMe drives and USB ports you can install.

This chapter explains chipsets, form factors and voltage regulator modules (VRMs), clarifies interfaces such as PCI Express and M.2, and provides a concise installation guide with real-world examples. By understanding these fundamentals you

can choose a board that matches your needs today and adapts to tomorrow's AI-driven workloads.

Chipsets: the platform's core

A chipset is a collection of controllers that augment the CPU with additional PCIe lanes, USB ports and other I/O. On AMD's AM5 platform, X670/X670E chipsets provide 44 PCIe lanes, support PCIe 5.0 for graphics and storage, and feature more high-speed USB and extra SATA ports. B650/B650E boards offer 36 lanes, typically limited to PCIe 4.0 and fewer USB ports.